Amirhossein Nazeri

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Professional Experience

Committee Member | *Standards Council of Canada* Dec 2025 – Present

- Contribute to international EV and grid-integration standards with IEC TC69 and ISO TC22
- Proposed users' safety framework for EV fast chargers with non-isolated power converters.

Student Researcher | *eLeapPower* collaboration with *University of Toronto* Jan 2022 – April 2025

- Developed and experimentally validated analytical models to predict common-mode currents in grid-connected EV DC fast chargers, enabling system-level safety assessment and grid compliance evaluation.
- Analyzed multi-limb coupled magnetic structures for multiphase DC EV chargers, reducing battery current ripple by 66% and improving charger efficiency.

Graduate Research Assistant | *University of Toronto* Sep 2021 – Present

- Developed a user safety framework for LV DC grid-connected equipment by integrating IEC, ISO, UL, and SAE standards.
- Designed and implemented a 60 kW grid-connected DC fast charging system using dual inverters, including real-time converter control, grid compliance verification, and system efficiency (97%).
- Developed an 800 V EV battery emulator compliant with SAE J1772 and IEC 61851-1 to enable system-level testing of grid-connected fast charging systems.

PhD Student | *University of Toronto* Sep 2021 – Present

- Modeled and analyzed a 1.5 MW grid-tied PV plant in PSCAD, including converter control, unbalanced grid operation, and transient stability.
- Simulated MV/LV distribution networks and protection schemes in DIgSILENT and PSCAD, including GE D60 relay coordination and COMTRADE-based validation.
- Designed and experimentally validated dq-frame control of a PMSM using a grid-connected 3φ VSC
- Built an EMT-based circuit simulator in MATALB, achieving <0.02% error in solutions.
- Designed and experimentally validated a PV micro-converter with variable-step P&O MPPT.

Research Intern | *University of Tehran* Jun 2019 – Sep 2020

- Developed a wireless monitoring and measurement network for microgrids.
- Created a software for remote microgrid management and data acquisition.

Relevant Skills

- **Power System Simulation:** PSCAD, DIgSILENT PowerFactory, PowerWorld, PSS/E, MATPOWER
- **Circuit Simulation:** PLECS, Altium Designer, NI Multisim, PSPICE, Proteus, HFSS, ADS, Sentaurus TCAD, Modelsim
- **Computation and Programming:** MATLAB and Simulink, Python, C++, C, Git, R, HTML, CSS
- **Hardware:** TI/dsPIC controllers, ARM STM32/AVR Microcontrollers, FPGA, Verilog, Quartus

Education

Ph.D in Electrical and Computer Engineering (Energy Systems) 2021 – Present

University of Toronto GPA: 3.86/4

Bachelor of Science in Electrical Engineering 2017 – 2021

University of Tehran GPA: 3.87/4